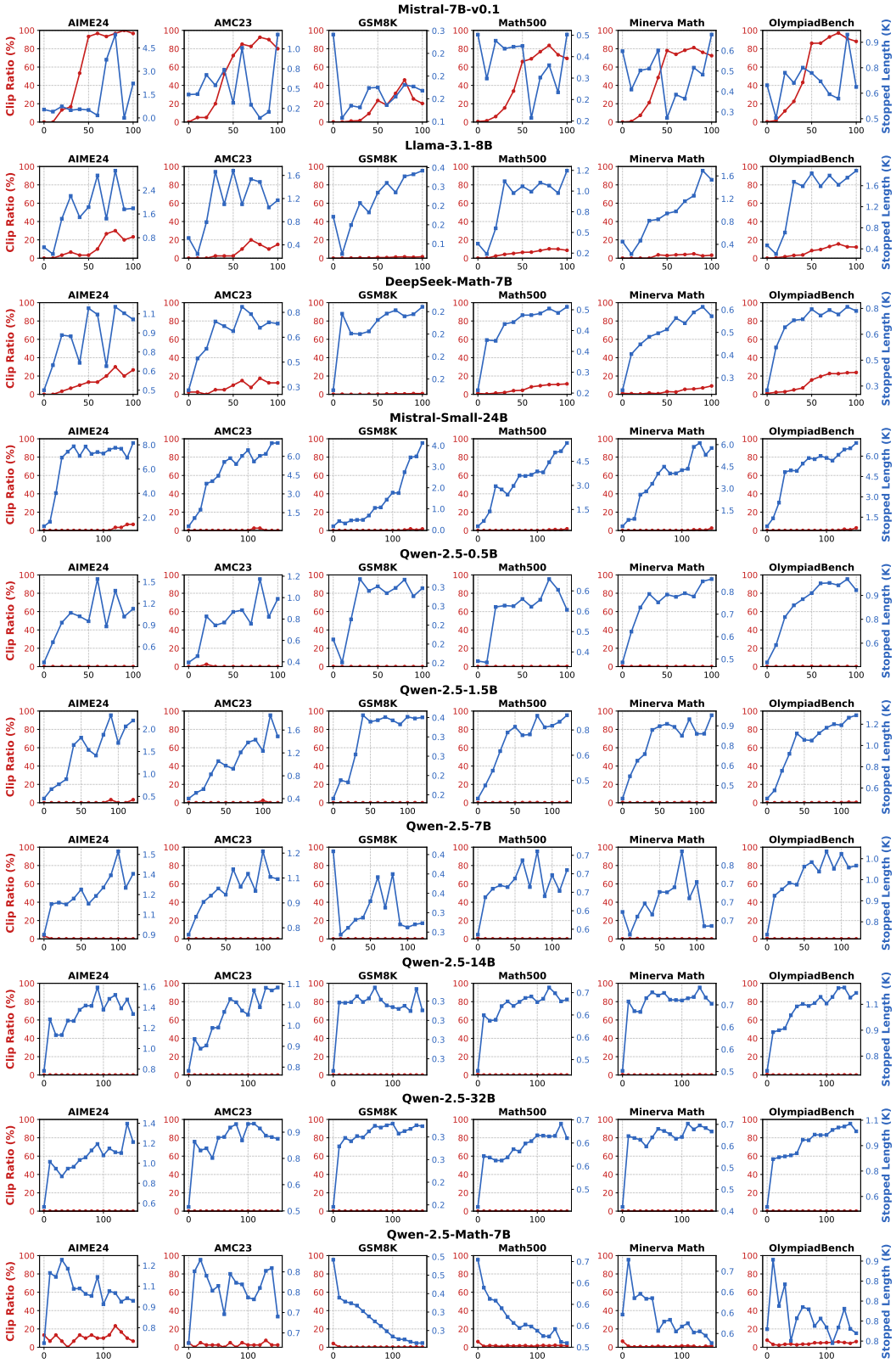


Figure 12: A detailed evaluation of clip ratio and stopped length throughout the training steps for various models. The x-axis represents the training steps, with the red line showing the clip ratio trend and the blue line depicting the average stopped length.

| Behavior | Score by GPT-4o (%) | Score by Human (%) | Raw Agreement (%) |
|---|---|---|---|
| Verification | 78.10% (82/105) | 85.71% (90/105) | 90.48% (95/105) |
| Backtracking | 33.33% (35/105) | 35.24% (37/105) | 98.10% (103/105) |
| Subgoal Setting | 66.67% (70/105) | 74.29% (78/105) | 90.48% (95/105) |
| Enumeration | 61.90% (65/105) | 63.81% (67/105) | 94.29% (99/105) |

Table 3: The consistency between GPT-4o labeled reasoning behaviors and human annotations

| Init Model | GSM8K | MATH 500 | Minerva Math | Olympiad Bench | AIME24 (pass@1) | AMC23 | Avg. |
|---|---|---|---|---|---|---|---|
| 0 Step | 92.0 | 70.6 | 36.8 | 36.6 | 16.7 | 45.0 | 49.6 |
| 10 Step | 93.0 | 69.4 | 39.7 | 32.3 | 10.4 | 44.1 | 48.2 |
| 20 Step | 92.6 | 65.2 | 34.2 | 30.7 | 6.7 | 38.4 | 44.6 |
| 200 Step | 90.3 | 59.0 | 31.6 | 23.3 | 2.1 | 26.9 | 38.9 |
| 1000 Step | 88.9 | 48.8 | 27.6 | 20.7 | 2.5 | 18.1 | 34.4 |
| 2000 Step | 89.8 | 49.0 | 23.2 | 18.1 | 0.8 | 20.3 | 33.5 |
| 4000 Step | 87.7 | 52.0 | 23.5 | 17.2 | 2.1 | 21.6 | 34.0 |

Table 4: Experimental results from multiple Mistral-Small-24B models, each fine-tuned with a different number of SFT steps on a general SFT dataset for RL. The "number of steps" refers to the number of SFT steps applied. The reported benchmarks reflect the performance metrics on various evaluation benchmarks, measured using the model that achieved the best average performance after 100 iterations of reinforcement learning training.

Our results indicate a generally good level of agreement between GPT-4o and human annotations. However, GPT-4o tends to be more conservative when labeling certain behaviors such as Verification and Subgoal Setting. Upon closer examination, we observe that in cases with long CoT containing multiple reasoning behaviors, the model often favors labeling more obvious behaviors like Enumeration, while overlooking subtler ones.

## F Impact of General SFT on the Performance of Reinforcement Learning

We also investigated the general SFT setting beyond math-related datasets. In this setup, we first conducted SFT on Mistral-Small-24B using the widely adopted OpenHermes-2.5 dataset.[3] We implement with LLaMA-Factory (Zheng et al., 2024) and adopt common hyperparameters of SFT, including 512 examples per batch with a constant learning rate of 1e-5. For consistency with our other experiments, we fine-tuned the model using the Qwen chat template. After SFT, we preserved multiple checkpoints at different training steps, and nearly 800 steps correspond to 1 epochs on the SFT dataset. We then performed reinforcement learning on these models using identical hyperparameters as in our zero-RL training experiments.

Table 4 presents our findings, with performance reported as the best results achieved during RL training up to 100 iterations. The results demonstrate an inverse relationship between SFT steps and subsequent RL performance: models with more SFT steps showed diminished performance after RL training. While the average performance after 10 SFT steps remained comparable to the base model, it still exhibited some negative effects. More significantly, models with more than 20 steps showed substantially reduced RL potential. Therefore, we conclude that RL training produces the best performance gain when applied directly to the base model without any supervised fine-tuning, i.e., the zero RL training.

[3] https://huggingface.co/datasets/teknium/OpenHermes-2.5

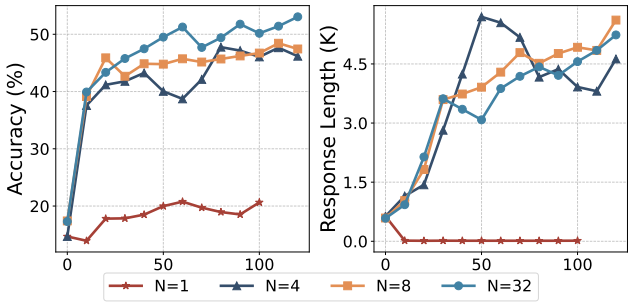


Figure 13: Comparison of accuracy and response length using different sampling numbers N = 1, 4, 8, 32. The training data is the Hard part (MATH lv.3–5) with the same setting in main results, as described in § 2.1.

## G Impact of Exploration-Related Hyperparameters

In this section, we examine the effects of exploration-related hyperparameters on "zero-training." Drawing inspiration from Zeng et al. (2025b); Liu et al. (2024), we focus on two key factors: sampling size (the number of responses per query) and sampling temperature.

**Sampling Size:** We examine how varying sampling sizes $N \in \{1, 4, 8, 16, 32\}$ influence the training process using the Mistral 24B model; these results are presented in Figure 13. Our analysis reveals a clear trend: as $N$ increases, the model's average performance notably improves, and variability in response lengths becomes significantly more stable. For example, after 100 training steps, the scenario with $N = 32$ achieves an average accuracy approximately 6 points higher than that with $N = 8$. Conversely, smaller sampling sizes ($N = 1$ and $N = 4$) cause training instability and potential collapse, indicated by rapid growth in generated length without corresponding accuracy improvements. We hypothesize that larger sample sizes enable the model to explore a broader and more diverse training space, which stabilizes advantage estimation and sustains continuous performance improvement.

**Sampling Temperature:** We conduct research on Qwen-2.5-0.5B to analyze the impact of sampling temperature during both training and evaluation on model performance. The results, presented in Figure 16 , indicate that training with higher temperatures generally leads to better average performance. For instance, models trained with temperatures of 1.0 and 1.2 outperform those trained with 0.8 and 0.6. Additionally, we find that the optimal evaluation temperature depends on the training temperature. Specifically, models trained at higher temperatures require higher sampling temperatures during evaluation, as using greedy sampling often results in repetitive outputs. Conversely, models trained at lower temperatures perform best when evaluated with lower sampling temperatures.

## H SimpleRL-Zoo For Qwen2.5-Math-7B

In this section, we conduct experiments on Qwen2.5-Math-7B (Yang et al., 2024a) using the "hard part" data, as described in § 2.1, which consists of only 8K examples from MATH lv3-5. We apply both the PPO and GRPO algorithms to train our base model, and the overall evaluation results across training steps are shown in Figure 14. The final performance and response length for both algorithms converge to similar values, with GRPO slightly outperforming PPO. While the performance continues to improve, the response length does not exhibit a similar trend. Specifically, the stopping length for both algorithms remains relatively unchanged, and fluctuations in the average response length are primarily attributed to changes in the clip ratio. There are two main reasons for this behavior: First, the maximum context length for Qwen2.5-Math-7B is 4K, which is limited compared to other models with context lengths exceeding 8K, leading to a high clip ratio. Second, as a math-specific model, Qwen2.5-Math-7B already performs very well on MATH, the dataset